Yeast Fermentation

Student worksheet | Flexible classroom investigation | Biology 2e, Chapter 7

1. Plan before you mix

Names: _____ Class/date: _____

Pair/team: _____ Pooled set ID: _____ Assigned flask IDs: _____

Question: How does the available sugar affect gas accumulation in a yeast culture? Your team prepares one or two independent flasks. Pool results with other teams to compare three no-added-sugar, three glucose and three sucrose trials.

A balloon can collect gas from a culture. Measuring its circumference means measuring the distance around its widest part. This gives a relative indicator of gas accumulation, not a volume in milliliters. Predict the results before starting.

Teacher-selected setup

Setting	Record your assignment
Method / measurement / unit	Balloon / circumference / cm OR: _____
Flask size / final liquid volume	_____ mL / _____ mL
Treatment stock / yeast suspension per flask	_____ mL / _____ mL
Sugar stock concentration / yeast stock concentration	_____ / _____
Temperature / reading interval / end time	_____ C / _____ min / _____ min

Prediction: Compare the three treatments and explain your reasoning.

What will you change? What will you measure? Name three things to keep the same.

Before beginning: confirm your flask IDs, stock labels, volumes and timing with the teacher. Wear eye protection, do not taste the mixtures, and never seal a culture with a rigid cap. Tell the teacher about damaged glassware or a balloon allergy.

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2. Prepare and record your own flasks

Label each flask. Add its assigned treatment stock. Ask for the yeast stock to be gently remixed immediately before measuring your aliquot. Add yeast last, mix consistently, promptly attach the balloon, and start that flask's timer. Use a fresh mixture for every flask.

At each elapsed time, gently measure the widest balloon circumference without squeezing. Record NI if not inflated, M if a reading is missing, and L with a note for a suspected leak. Do not substitute zero for NI or M. For gas collection or sensors, use the teacher's equipment procedure and the correct units.

Measurement: _____ Unit: _____ Pooled set ID: _____

Flask A ID: ____ Treatment: _____ Start time: ____ Temperature: ____ C

Flask B ID: ____ Treatment: _____ Start time: ____ Temperature: ____ C

Elapsed min*	Flask A reading	Flask B reading	Notes: foam, leaks, delays, changes
0			
5			
10			
15			
20			
25			
30			

*Use 0-30 minutes at 5-minute intervals unless your teacher assigns another schedule; relabel times. Read each flask at its own elapsed time. Leave the second flask column unused if you have only one.

Time to first visible inflation (if observed): A _____ min; B _____ min

Describe anything that could affect your measurements. Keep the raw readings.

Share your flask IDs, raw readings, units, temperature and notes with your pooled group. Nine flasks are nine independent trials; seven readings from one flask are still only one trial.

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3. Pool three trials of each treatment

Pooled set ID: _____ Measurement: _____ Unit: _____

Pool only matching recipes, flask sizes, equipment and temperatures. Keep virtual mL results separate from balloon cm or sensor readings. Copy actual times if your schedule differs. NI = not inflated; M = missing; L = suspected leak.

No added sugar

Elapsed min	N1	N2	N3	Numeric mean	n / NI count
0					
5					
10					
15					
20					
25					
30					

Glucose

Elapsed min	G1	G2	G3	Numeric mean	n / NI count
0					
5					
10					
15					
20					
25					
30					

Sucrose

Elapsed min	S1	S2	S3	Numeric mean	n / NI count
0					
5					
10					
15					
20					
25					
30					

Mean = sum of comparable numeric readings / number of those readings (n). Report NI separately. If n is below 3, label the mean as numeric-only; it does not represent all three flasks. Keep flagged observations and explain any exclusion.

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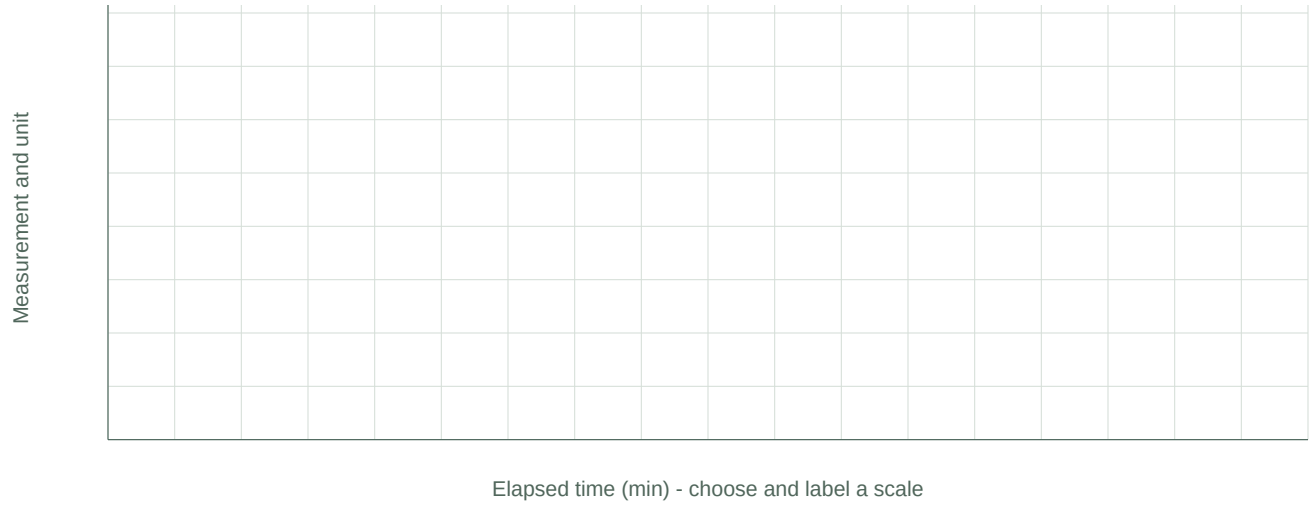
4. Compare evidence and explain

Graph measurement versus elapsed time. Label scales and units; use different colors or symbols and a legend for the three treatments. Mark missing/NI readings as gaps, not zero. Label means with n when fewer than three trials contribute.



Legend: No sugar _____ Glucose _____ Sucrose _____

1. What do the data support? Compare treatments using numbers, units and variation among the three independent trials.

2. Why include no-added-sugar yeast? Would sugar solution without yeast answer a different question?

3. Explain why balloon size alone cannot measure ATP production or prove that oxygen was absent. Identify a measurement limitation.

After the comparison: propose one follow-up test (for example sugar concentration or temperature). State what you would change and keep constant. Get teacher approval before testing; record new results separately.